

SRI GURU NANAK PUBLIC SCHOOL



SESSION: 2025-2026

A Project Report On **E-COMMERCE**

For CBSE 2026 Examination
(As a part of the Computer Science Course(083))

SUBMITTED BY:

YATHARTH JINDAL

CLASS-XII A

ROLL NO:

UNDER THE GUIDANCE OF:

Mrs.Honey Kaur

(PGT Computer Science)

Certificate

This is to certify that ***YATHARTH JINDAL*** of class XII A of SRI GURU NANAK PUBLIC SCHOOL has done his project on ***E-COMMERCE*** under my supervision. He has taken her interest and has shown at most sincerity in completion of this project. I certified this project upto my expectation and as per guidelines issued by **CBSE, NEW DELHI**

INTERNAL EXAMINER
Mrs. Honey Kaur

EXTERNAL EXAMINER

PRINCIPAL
Mrs. Shikha Gupta

Acknowledgement

It is with pleasure that I acknowledge my sincere gratitude to our teacher, **MRS. HONEY KAUR** who taught and undertook the responsibility of teaching the subject computer science. I have been greatly benefited from his classes.

I am especially indebted to our principal **MRS. SHIKHA GUPTA** who has always been a source of encouragement and support and without whose inspiration this project would not have been a successful I would like to place record heartfelt thanks to him.

Finally, I would like to express my sincere appreciation for all the other students for my batch their friendship & the fine time that we all share together.

NAME: YATHARTH JINDAL
CLASS: XII-A
SIGN :

Hardware And Software Required

Hardware

1. Desktop Monitor: 15 inch,
2. Mouse :
3. Keyboard:

Software

1. My SQL (version 8.0)
2. Python (version 3.13.1)
3. Python Connector Module
4. Operating system: Windows 11
5. MS-Word 2019

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ROLE OF TEAM MEMBERS

S.NO.	NAME	ROLES
01	YATHARTH JINDAL	CODING
02	NIPUN GOEL	EXCECUTION
03	SAHIBJEET SINGH	DATA COLLECTION

INTRODUCTION

The rapid growth of e-commerce has transformed the way businesses and consumers interact by enabling digital transactions, efficient order management, and seamless communication between buyers and sellers. E-commerce systems provide a structured platform where products can be listed, orders can be placed, and feedback can be managed securely.

This project focuses on the design and implementation of a console-based E-Commerce Management System using Python and MySQL. The system provides essential e-commerce functionalities such as customer and seller registration, product management, order placement, order history tracking, and feedback management. The project emphasizes backend logic, database connectivity, and data integrity rather than graphical user interfaces.

The system allows customers to browse products, place orders, view order history, and submit feedback. At the same time, sellers can add and manage products, view received orders, and store seller feedback securely. All user data, product details, orders, and feedback information are stored and managed using a MySQL relational database, ensuring data consistency and reliability.

Python is used as the backend programming language to implement application logic, while MySQL serves as the database for storing and retrieving information efficiently. This project demonstrates practical implementation of database connectivity, CRUD operations, and role-based functionality in an e-commerce environment.

The primary objective of this project is to develop a secure, structured, and user-friendly e-commerce system that showcases the integration of Python programming with MySQL database management in a real-world application scenario.

MYSQL DATABASE AND TABLES

E-COMMERCE DATABASE:

```
mysql> use ecommerce;
Database changed
mysql> show tables;
+-----+
| Tables_in_ecommerce |
+-----+
| customers            |
| feedbacks            |
| orders               |
| products              |
| seller                |
| sfeedback             |
+-----+
6 rows in set (0.23 sec)
```

CUSTOMER TABLE:

```
mysql> SELECT * FROM CUSTOMERS;
+-----+-----+-----+-----+-----+-----+-----+
| Customer_ID | User_Name | Password | Cust_Name | Address | Phone_No | Email_ID |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | KANNU | YJ123 | YATHARTH JINDAL | U-23 | 9889977665 | yj@gmail.com |
| 2 | Nipun05 | 12345678 | Nipun Goel | A-23 | 9899922334 | nipungoel@gmail.com |
| 3 | SS22 | 1133355555 | Sahibjeet Singh | A-09, MAJLIS PARK | 9887755441 | sahibsingh23@gmail.com |
+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

FEEDBACK TABLE:

```
mysql> SELECT * FROM FEEDBACKS;
```

Customer_ID	Feedback	Seller_ID	Product_ID	Seller_Name
1	Good book	1	101	Alakh Pandey
2	Bad service	1	102	Alakh Pandey
3	GOOD	1	102	Alakh Pandey
3	GOOD	2	111	Udit Singh

4 rows in set (0.00 sec)

ORDER TABLE:

```
mysql> SELECT * FROM ORDERS;
```

Transaction_ID	Customer_ID	Product_ID	Date_of_Dispatch	Quantity	Price	Amount
1	1	101	2026-01-04	25	349.00	8725.00
2	2	102	2026-01-05	5	349.00	1745.00
3	2	101	2026-01-05	10	349.00	3490.00
4	2	111	2026-01-05	25	250.00	6250.00
5	3	111	2026-01-05	2	250.00	500.00
6	3	102	2026-01-05	25	349.00	8725.00
7	3	101	2026-01-05	2	349.00	698.00

7 rows in set (0.00 sec)

PRODUCT TABLE:

```
mysql> SELECT * FROM PRODUCTS;
```

Product_ID	Category	Name	Quantity	Price	Brand_Name	Seller_ID
101	JEE BOOKS	HC VERMA PHYSICS VOL 1	15	349.00	PW	1
102	JEE BOOKS	HC VERMA PHYSICS VOL 2	45	349.00	PW	1
103	JEE BOOKS	CHEMISTRY FOR JEE MAINS PYQ 2019-2024	35	149.00	PW	1
111	BOOKS	NEET CHEMISTRY PYQS 2019-2025	25	250.00	ALLEN	2

4 rows in set (0.00 sec)

SELLER TABLE:

```
mysql> SELECT * FROM SELLER;
```

Seller_ID	Seller_Name	Brand_Name	User_Name	Password	Phone_No	Email_ID
1	Alakh Pandey	PW	PW123	122333	8900889976	pw123@gmail.com
2	Usit Singh	ALLEN	ALLEN22	123	8989898966	usitallen12@gmail.com

```
2 rows in set (0.00 sec)
```

SELLER'S FEEDBACK TABLE:

```
mysql> SELECT * FROM SFEEDBACK;
```

Seller_ID	Feedback	Rating
1	Good	4
1	GOOD	3
2	Average	3

```
3 rows in set (0.01 sec)
```

CODING

project code.py - D:\Yatharth 12A\project code.py (3.12.3)

File Edit Format Run Options Window Help

```
import mysql.connector
from datetime import date

# -----
# DATABASE CONNECTION
# -----
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="tiger",
    database="ecommerce"
)
cursor = conn.cursor()

# -----
# CHECK USERNAME EXISTS
# -----
def check_username(table, username):
    query = "SELECT Username FROM " + table + " WHERE Username=%s"
    cursor.execute(query, (username,))
    return cursor.fetchone()

# -----
# CUSTOMER REGISTRATION
# -----
def create_customer():
    print("\n---- Customer Registration ----")
    fname = input("Full Name: ")
    uname = input("Username: ")
    while check_username("customers", uname):
        print("\u2604 Username already exists! Try another.")
        uname = input("Username: ")

    pwd = input("Password: ")

    while True:
        phone = input("Phone (10 digits): ")
        if phone.isdigit() and len(phone) == 10:
            break
        else:
            print("\u2604 Phone number must be exactly 10 digits")

    address = input("Address: ")
    email = input("Email: ")

    cursor.execute("SELECT MAX(Customer_ID) FROM customers")
    r = cursor.fetchone()
    cid = 1 if r[0] is None else r[0] + 1

    cursor.execute("""
        INSERT INTO customers
        (Customer_ID, Cust_Name, User_Name, Password, Phone_No, Address, Email_ID)
        VALUES (%s,%s,%s,%s,%s,%s,%s)
        """, (cid, fname, uname, pwd, phone, address, email))

    conn.commit()
    print(f"\u2705 Registered Successfully! Customer ID: {cid}")

# -----
# CUSTOMER LOGIN
# -----
def login_customer():
    uname = input("Username: ")
    pwd = input("Password: ")

    cursor.execute("""
        SELECT Customer_ID, Cust_Name
        FROM customers
        WHERE User_Name=%s AND Password=%s
        """, (uname, pwd))

    u = cursor.fetchone()
    if u:
        print(f"\u2705 Welcome {u[1]}")
        customer_menu(u[0])
    else:
        print("\u2604 Invalid Login")
```

```

# -----
# SELLER REGISTRATION
# -----
def create_seller():
    print("\n--- Seller Registration ---")
    sname = input("Seller Name: ")
    bname = input("Brand Name: ")
    uname = input("Username: ")
    while check_username("seller", uname):
        print("\u2609 Username already exists!")
        uname = input("Username: ")

    pwd = input("Password: ")

    while True:
        phone = input("Phone (10 digits): ")
        if phone.isdigit() and len(phone) == 10:
            break
        else:
            print("\u274c Phone number must be exactly 10 digits")

    email = input("Email: ")

    cursor.execute("SELECT MAX(Seller_ID) FROM seller")
    r = cursor.fetchone()
    sid = 1 if r[0] is None else r[0] + 1

    cursor.execute("""
        INSERT INTO seller
        (Seller_ID, Seller_Name, Brand_Name, Username, Password, Phone_No, Email_ID)
        VALUES (%s,%s,%s,%s,%s,%s,%s)
    """, (sid, sname, bname, uname, pwd, phone, email))

    conn.commit()
    print(f"\u2705 Seller Registered! Seller ID: {sid}")

# -----
# SELLER LOGIN
# -----
def login_seller():
    uname = input("Username: ")
    pwd = input("Password: ")

    cursor.execute("""
        SELECT Seller_ID, Seller_Name
        FROM seller
        WHERE UserName=%s AND Password=%s
    """, (uname, pwd))

    s = cursor.fetchone()
    if s:
        print(f"\u2705 Welcome Seller {s[1]}")
        seller_menu(s[0])
    else:
        print("\u274c Invalid Login")

# -----
# VIEW PRODUCTS
# -----
def search_products():
    cursor.execute("""
        SELECT Product_ID, Category, Name, Quantity, Price, Brand_Name
        FROM products
    """)
    print("\n--- Available Products ---")
    for r in cursor.fetchall():
        print(r)

# -----
# VIEW ORDER HISTORY (CUSTOMER)
# -----
def view_history(cid):
    cursor.execute("""
        SELECT orders.Transaction_ID,
        products.Name,
        orders.Quantity,
        orders.Date_of_Dispatch,
        orders.Amount
        FROM orders
        JOIN products ON orders.Product_ID = products.Product_ID
        WHERE orders.Customer_ID = %s
    """, (cid,))

    rows = cursor.fetchall()
    if not rows:
        print("\u2609 No orders found")
        return

    print("\n--- Your Order History ---")
    for r in rows:
        print(r)

```

```

# -----
# PLACE ORDER
# -----
def place_order(cid):
    pid = int(input("Product ID: "))
    qty = int(input("Quantity: "))

    cursor.execute("SELECT Quantity, Price FROM products WHERE Product_ID=%s", (pid,))
    r = cursor.fetchone()
    if not r:
        print("❌ Invalid Product")
        return

    stock, price = r
    if qty > stock:
        print("⚠️ Not enough stock")
        return

    total = qty * price

    cursor.execute(
        "UPDATE products SET Quantity=Quantity-%s WHERE Product_ID=%s",
        (qty, pid)
    )

    cursor.execute("SELECT MAX(Transaction_ID) FROM orders")
    r = cursor.fetchone()
    tid = 1 if r[0] is None else r[0] + 1

    cursor.execute("""
        INSERT INTO orders
        (Transaction_ID, Customer_ID, Product_ID, Date_of_Dispatch, Quantity, Price, Amount)
        VALUES (%s,%s,%s,%s,%s,%s,%s)
        """, (tid, cid, pid, date.today(), qty, price, total))

    conn.commit()
    print("✅ Order Placed Successfully")

# -----
# CUSTOMER FEEDBACK
# -----
def customer_feedback(cid):
    pid = int(input("Product ID: "))
    sid = int(input("Seller ID: "))
    sname = input("Seller Name: ")
    fb = input("Feedback: ")

    cursor.execute("""
        INSERT INTO feedbacks
        (Customer_ID, Feedback, Seller_ID, Product_ID, Seller_Name)
        VALUES (%s,%s,%s,%s,%s)
        """, (cid, fb, sid, pid, sname))

    conn.commit()
    print("✅ Feedback Submitted")

# -----
# SELLER FEEDBACK
# -----
def seller_feedback(sid):
    fb = input("Enter your feedback about customers/products: ")

    while True:
        rating = input("Rating (1-5): ")
        if rating.isdigit() and 1 <= int(rating) <= 5:
            rating = int(rating)
            break
        else:
            print("❌ Rating must be between 1 and 5 only")

    cursor.execute("""
        INSERT INTO sfeedback
        (Seller_ID, Feedback, Rating)
        VALUES (%s,%s,%s)
        """, (sid, fb, rating))

    conn.commit()
    print("✅ Seller Feedback Stored Successfully")

```

```

# -----
# SELLER MENU
# -----
def seller_menu(sid):
    while True:
        print("\n--- Seller Menu ---")
        print("1. View My Products")
        print("2. Add Product")
        print("3. View Orders")
        print("4. Give Seller Feedback")
        print("5. Logout")

        c = input("Choice: ")
        if c == '1':
            view_seller_products(sid)
        elif c == '2':
            add_product(sid)
        elif c == '3':
            seller_orders(sid)
        elif c == '4':
            seller_feedback(sid)
        elif c == '5':
            break

def view_seller_products(sid):
    cursor.execute("SELECT * FROM products WHERE Seller_ID=%s", (sid,))
    for r in cursor.fetchall():
        print(r)

def seller_orders(sid):
    cursor.execute("""
        SELECT orders.Transaction_ID, products.Name,
               orders.Quantity, orders.Date_of_Dispatch, orders.Amount
        FROM orders
        JOIN products ON orders.Product_ID = products.Product_ID
        WHERE products.Seller_ID=%s
    """, (sid,))
    for r in cursor.fetchall():
        print(r)

def add_product(sid):
    pid = int(input("Product ID: "))
    cat = input("Category: ")
    name = input("Name: ")
    qty = int(input("Quantity: "))
    price = float(input("Price: "))

    cursor.execute("SELECT Brand_Name FROM seller WHERE Seller_ID=%s", (sid,))
    bname = cursor.fetchone()[0]

    cursor.execute("""
        INSERT INTO products
        (Product_ID, Category, Name, Quantity, Price, Brand_Name, Seller_ID)
        VALUES (%s,%s,%s,%s,%s,%s,%s)
    """, (pid, cat, name, qty, price, bname, sid))

    conn.commit()
    print("✔ Product Added")

```

```

# -----
# CUSTOMER MENU
# -----
def customer_menu(cid):
    while True:
        print("\n--- Customer Menu ---")
        print("1. View Products")
        print("2. Place Order")
        print("3. View Order History")
        print("4. Give Feedback")
        print("5. Logout")

        c = input("Choice: ")
        if c == '1':
            search_products()
        elif c == '2':
            place_order(cid)
        elif c == '3':
            view_history(cid)
        elif c == '4':
            customer_feedback(cid)
        elif c == '5':
            break

```

```

# -----
# MAIN MENU
# -----
def main_menu():
    while True:
        print("\n=== E-Commerce System ===")
        print("1. Customer Register")
        print("2. Customer Login")
        print("3. Seller Register")
        print("4. Seller Login")
        print("5. Exit")

        ch = input("Choice: ")
        if ch == '1':
            create_customer()
        elif ch == '2':
            login_customer()
        elif ch == '3':
            create_seller()
        elif ch == '4':
            login_seller()
        elif ch == '5':
            break

main_menu()
cursor.close()
conn.close()

```

CUSTOMER OUTPUTS

STARTING INTERFACE:

```
===== RESTART: D:\Yatharth 12A\project code.py =====  
  
=== E-Commerce System ===  
1. Customer Register  
2. Customer Login  
3. Seller Register  
4. Seller Login  
5. Exit  
Choice: 1
```

CUSTOMER SIGNUP:

```
---- Customer Registration ----  
Full Name: Nipun Goel  
Username: Nipun05  
Password: 12345678  
Phone (10 digits): 9899922334  
Address: A-23  
Email: nipungoel@gmail.com  
✓ Registered Successfully! Customer ID: 2
```

CUSTOMER LOGIN:

```
=== E-Commerce System ===  
1. Customer Register  
2. Customer Login  
3. Seller Register  
4. Seller Login  
5. Exit  
Choice: 2  
Username: Nipun05  
Password: 12345678  
✓ Welcome Nipun Goel
```

CUSTOMER MAIN-MENU:

---- Customer Menu ----

1. View Products
2. Place Order
3. View Order History
4. Give Feedback
5. Logout

Choice: 1

VIEWING PRODUCTS:

---- Customer Menu ----

1. View Products
2. Place Order
3. View Order History
4. Give Feedback
5. Logout

Choice: 1

--- Available Products ---

(101, 'JEE BOOKS', 'HC VERMA PHYSICS VOL 1', 25, Decimal('349.00'), 'PW')
(102, 'JEE BOOKS', 'HC VERMA PHYSICS VOL 2', 50, Decimal('349.00'), 'PW')
(103, 'JEE BOOKS', 'CHEMISTRY FOR JEE MAINS PYQ 2019-2024', 35, Decimal('149.00'), 'PW')

PLACING ORDER:

---- Customer Menu ----

1. View Products
2. Place Order
3. View Order History
4. Give Feedback
5. Logout

Choice: 2

Product ID: 102

Quantity: 5

✓ Order Placed Successfully

---- Customer Menu ----

1. View Products
2. Place Order
3. View Order History
4. Give Feedback
5. Logout

Choice: 2

Product ID: 101

Quantity: 10

✓ Order Placed Successfully

VIEWING ORDER HISTORY:

----- Customer Menu -----

1. View Products
2. Place Order
3. View Order History
4. Give Feedback
5. Logout

Choice: 3

--- Your Order History ---

```
(2, 'HC VERMA PHYSICS VOL 2', 5, datetime.date(2026, 1, 5), Decimal('1745.00'))
(3, 'HC VERMA PHYSICS VOL 1', 10, datetime.date(2026, 1, 5), Decimal('3490.00'))
(4, 'NEET CHEMISTRY PYQS 2019-2025', 25, datetime.date(2026, 1, 5), Decimal('6250.00'))
```

TO GIVE FEEDBACK:

----- Customer Menu -----

1. View Products
2. Place Order
3. View Order History
4. Give Feedback
5. Logout

Choice: 4

Product ID: 102

Seller ID: 101

Seller Name: Alakh Pandey

Feedback: Bad service

✓ Feedback Submitted

LOGGING OUT:

----- Customer Menu -----

1. View Products
2. Place Order
3. View Order History
4. Give Feedback
5. Logout

Choice: 5

=== E-Commerce System ===

1. Customer Register
2. Customer Login
3. Seller Register
4. Seller Login
5. Exit

Choice: |

SELLER OUTPUTS

STARTING INTERFACE:

```
===== RESTART: D:\Yatharth 12A\project code.py =====  
=== E-Commerce System ===  
1. Customer Register  
2. Customer Login  
3. Seller Register  
4. Seller Login  
5. Exit  
Choice: 3
```

SELLER'S SIGNUP:

```
---- Seller Registration ----  
Seller Name: Usit Singh  
Brand Name: ALLEN  
Username: ALLEN22  
Password: 123  
Phone (10 digits): 8989898966  
Email: usitallen12@gmail.com  
✓ Seller Registered! Seller ID: 2
```

SELLER LOGIN:

```
=== E-Commerce System ===  
1. Customer Register  
2. Customer Login  
3. Seller Register  
4. Seller Login  
5. Exit  
Choice: 4  
Username: ALLEN22  
Password: 123  
✓ Welcome Seller Usit Singh
```

SELLER MAIN MENU:

```
---- Seller Menu ----  
1. View My Products  
2. Add Product  
3. View Orders  
4. Give Seller Feedback  
5. Logout  
Choice: 1
```

VIEWING PRODUCTS:

```
---- Seller Menu ----  
1. View My Products  
2. Add Product  
3. View Orders  
4. Give Seller Feedback  
5. Logout  
Choice: 1  
(111, 'BOOKS', 'NEET CHEMISTRY PYQS 2019-2025', 25, Decimal('250.00'), 'ALLEN', 2)
```

ADDING PRODUCTS:

```
----- Seller Menu -----  
1. View My Products  
2. Add Product  
3. View Orders  
4. Give Seller Feedback  
5. Logout  
Choice: 2  
Product ID: 111  
Category: BOOKS  
Name: NEET CHEMISTRY PYQS 2019-2025  
Quantity: 50  
Price: 250  
✓ Product Added
```

VIEWING ORDERS:

```
----- Seller Menu -----  
1. View My Products  
2. Add Product  
3. View Orders  
4. Give Seller Feedback  
5. Logout  
Choice: 3  
(4, 'NEET CHEMISTRY PYQS 2019-2025', 25, datetime.date(2026, 1, 5), Decimal('6250.00'))
```

GIVING FEEDBACK AS SELLER:

```
----- Seller Menu -----  
1. View My Products  
2. Add Product  
3. View Orders  
4. Give Seller Feedback  
5. Logout  
Choice: 4  
Enter your feedback about customers/products: Average  
Rating (1-5): 3  
✓ Seller Feedback Stored Successfully
```

LOGGING OUT:

```
----- Customer Menu -----  
1. View Products  
2. Place Order  
3. View Order History  
4. Give Feedback  
5. Logout  
Choice: 5  
  
=== E-Commerce System ===  
1. Customer Register  
2. Customer Login  
3. Seller Register  
4. Seller Login  
5. Exit
```

EXIT:

```
=== E-Commerce System ===
```

```
1. Customer Register
```

```
2. Customer Login
```

```
3. Seller Register
```

```
4. Seller Login
```

```
5. Exit
```

```
Choice: 5
```

```
|
```

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